

Finance Learning Site

Project Blueprint — full specification for build, design, and long-term content workflow

A self-built, ever-growing reference for general finance knowledge — notes, diagrams, MCQs, and flashcards, added one topic at a time.

1. Project Overview

A personal, long-term learning site covering general finance fundamentals — not CFA-exam-specific, but broad enough to support CFA study while staying useful as a standalone knowledge base. Built and expanded topic by topic, on a daily/weekly cadence, so it doubles as a habit-building project (steady commits) and an actual study tool.

Goals, in priority order

- **Low friction to add a new topic** — a single command scaffolds everything needed for a day's contribution.
- **Genuinely useful learning tools per topic** — notes, a diagram, MCQs, and flashcards, not just walls of text.
- **Polished, considered UI** — should feel like a real, deliberate product, not a template or a bare markdown dump.

Topic areas (illustrative, not exhaustive)

- Corporate Finance — time value of money, capital budgeting, cost of capital
- Financial Statements — income statement, balance sheet, cash flow statement, ratio analysis
- Equities — valuation methods, market structure, equity risk
- Fixed Income — bond pricing, yield curves, duration & convexity
- Derivatives — forwards, futures, options, swaps
- Portfolio Management — risk/return, diversification, CAPM, portfolio construction
- Economics — micro/macro fundamentals relevant to markets

2. Tech Stack

Layer	Technology	Why
Framework	Astro	Content-first, ships minimal JS, great for a reading-heavy site with occasional interactive islands
Interactivity	React (islands)	Used only for quiz, flashcards, and animated UI pieces — not the whole site
Styling	Tailwind CSS	Fast to iterate; custom theme config, not defaults
Animation	Framer Motion	Hover, scroll-reveal, and flip-card interactions
Diagrams	Mermaid	Text-based, diffable in git, renders directly from .mmd files

Layer	Technology	Why
Content	Astro Content Collections	Type-safe (Zod schema) markdown/MDX + JSON per topic
Navigation	Astro View Transitions	Smooth cross-page feel between topic pages
Hosting	Vercel	Clean Astro deploys, easy to keep live and shareable

3. Content Model

Each topic is a self-contained folder. This is the core unit of the whole project — everything else (pages, CLI, schema) exists to serve this structure.

```
content/topics/{topic-slug}/
  note.mdx           notes, written in my own words
  diagram.mmd        Mermaid diagram illustrating the concept
  quiz.json           5-10 MCQs: question, options[], correctIndex, explanation
  flashcards.json     question/answer pairs: front, back
  (frontmatter/meta) title, category, difficulty, dateAdded, tags[]
```

Defined as an Astro content collection with a Zod schema, so a malformed topic file fails the build loudly instead of silently breaking the site.

Categories (frontmatter field)

- Corporate Finance
- Financial Statements
- Equities
- Fixed Income
- Derivatives
- Portfolio Management
- Economics

4. Site Pages

Homepage

- Short hero intro with an original headline
- Browsable grid of topic cards, grouped by category
- Each card: muted category label, title, one-line description, progress indicator once studied
- Subtle "New" badge on the most recently added topic
- Simple search / filter by tag or category

Topic Page

- Centered, readable note column (max ~65-75ch line length)

- Mermaid diagram inline, in a clean bordered card
- Quiz + flashcard widgets as visually distinct "practice modules" below the note

Other Pages

- Category browse page
- All-topics list page

5. Design System

Inspiration: clean, editorial modern SaaS/edtech products (general UI language — whitespace, restrained palette, clean cards, subtle motion, flip-card pattern) — an original interpretation, not a clone of any specific product's branding or assets.

Color & Layout

- Light background by default (warm/cool near-white, not stark white); dark mode as a toggle, persisted in localStorage, with properly tuned contrast — not inverted colors
- One accent color used sparingly and consistently — deep blue or teal, calm and trustworthy, never flashy
- Restrained neutral grayscale everywhere else
- Generous whitespace — reading-first product, not a dense dashboard
- Cards: subtle 1px low-contrast borders or soft shadows, generous but not overly rounded corners

Typography

- Primary font: modern geometric/grotesk sans (Geist Sans or General Sans), self-hosted via @fontsource — not Google Fonts CDN
- Note body: same font, larger + more line-height, or paired with a serif (e.g. Source Serif 4) for an editorial reading feel
- Proper type scale in Tailwind config — distinct sizes/weights/tracking for hero, section headings, card titles, body, meta text
- Headlines: tighter tracking, heavier weight

Component Behavior

- **Flashcard widget** — centered card, real 3D flip (transform-style: preserve-3d, backface-visibility: hidden) via Framer Motion rotateY, ~400-500ms ease-out, click or spacebar, subtle scale/shadow pulse on click, muted "Didn't know / Knew it" buttons, progress indicator + prev/next
- **Quiz widget** — one question at a time, clean radio options, animated correct/incorrect highlight with scale pulse, explanation fades + slides in, smooth progress bar, summary screen with missed-question review
- **Navigation** — text logo, Topics / Categories / About; transparent over hero, gains backdrop-blur + background + soft shadow on scroll

6. Motion System

- Framer Motion for React islands; `whileInView` (or lightweight `IntersectionObserver`) for scroll reveals on static content
- Hero entrance: headline / subtext / CTA fade + slide up, staggered ~80-120ms apart
- Scroll reveals: cards/sections fade + slide up on enter (threshold ~0.2, once), staggered ~50-80ms per grid row
- Card hover: lift -2 to -4px + increased shadow, ~150-200ms ease
- Buttons: hover scale 1.0 → 1.02, smooth background/shadow transition
- Cross-page navigation via Astro View Transitions
- Shared lib/motion.ts with named duration/easing/stagger constants — consistent timing site-wide; smooth deceleration curve, e.g. cubic-bezier(0.16, 1, 0.3, 1)
- Respect prefers-reduced-motion — required, not optional
- Bias toward subtlety: "feels expensive and considered," never flashy

7. Daily Content Workflow

This is the part that determines whether the project actually gets used long-term. A single CLI command should produce everything needed for one day's contribution.

```
npm run new-topic

-> prompts for topic title + category
-> generates the slug
-> scaffolds note.mdx, quiz.json, flashcards.json (with example entries),
    and a placeholder diagram.mmd
-> prints file paths to fill in
```

Daily loop: run the command, write the note, draw the diagram, add a handful of MCQs/flashcards, commit, push. That's a day's contribution — and a commit for the streak.

8. Seed Content

One fully worked example topic, built end-to-end, to serve as the reference pattern for every topic after it.

Field	Value
Title	Time Value of Money
Category	Corporate Finance
Note	PV / FV / annuities, explained in full
Diagram	Cash flow timeline (Mermaid)
Quiz	5 real MCQs
Flashcards	5 real front/back pairs

9. Build Sequence

The full build is split into stages so each can be reviewed before moving on.

Stage	Scope
1 — Scaffold	Astro + content collections + Zod schema, new-topic CLI, seed "Time Value of Money" topic, basic unstyled pages working end-to-end
2 — Design system	Full Tailwind theme, typography, homepage/topic page layout, flashcard + quiz component visuals
3 — Motion layer	Framer Motion setup, scroll reveals, hover states, flip animation, View Transitions, reduced-motion support
4 — Ongoing	New topics added via the CLI, daily/weekly

10. Deployment & Documentation

- Deploy via Vercel — clean Astro support, live shareable URL
- README covering: project purpose, content folder structure, and exact steps to add a new topic (doubles as documentation for future-me)

This blueprint is the reference document for the Antigravity build prompts. Each build-sequence stage above corresponds to one prompt.